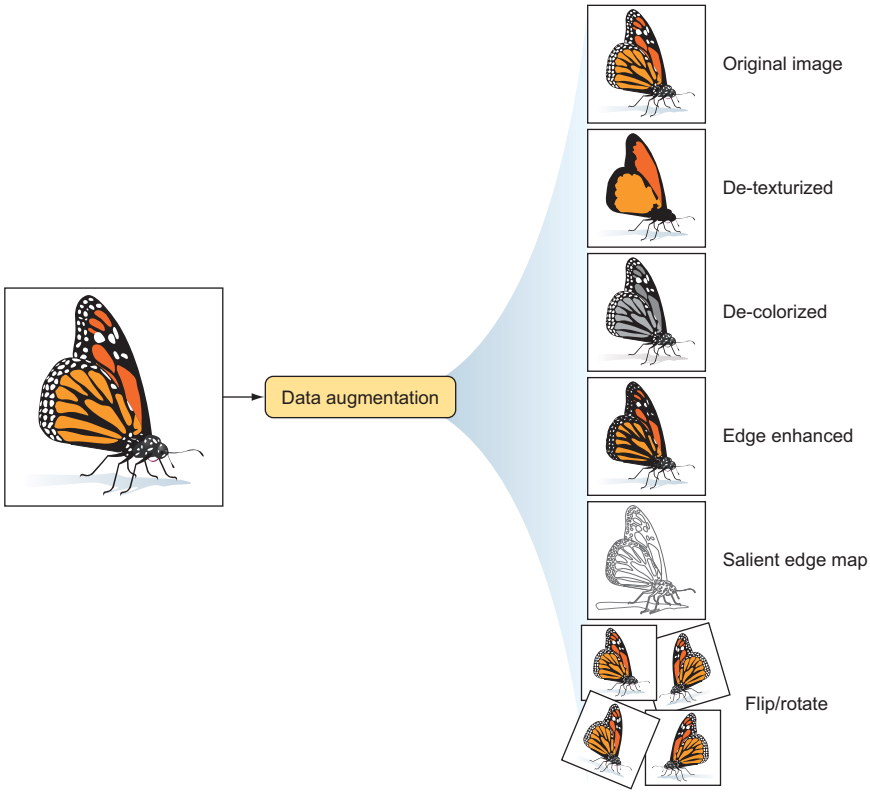

Figure 1.21 Image-augmentation techniques create modified versions of the input image to provide more examples for the ML model to learn from.

the appropriate processing techniques based on the dataset at hand and the problem you are solving. You will see many image-processing techniques throughout this book, helping you build your intuition of which ones you need when working on your own projects.

No free lunch theorem

This is a phrase that was introduced by David Wolpert and William Macready in "No Free Lunch Theorems for Optimizations" (*IEEE Transactions on Evolutionary Computation* 1, 67). You will often hear this said when a team is working on an ML project. It means that no one prescribed recipe fits all models. When working on ML projects, you will need to make many choices like building your neural network architecture, tuning hyperparameters, and applying the appropriate data preprocessing techniques. While there are some rule-of-thumb approaches to tackle certain problems, there is really no single recipe that is guaranteed to work well in all situations.